Control of Mixed-Autonomy Traffic via Autonomous Vehicles with Lane-Changing Behavior

From Reinforcement Learning to Interpretable Pair-aligned Controller

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Stop-and-go waves degrade traffic safety and efficiency

Phantom traffic jams

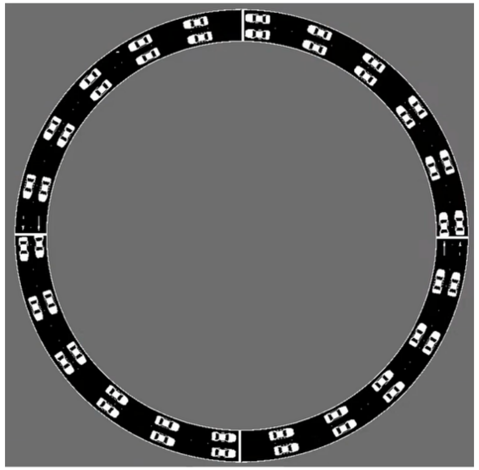
Random human braking introduces small disturbances into traffic flow.

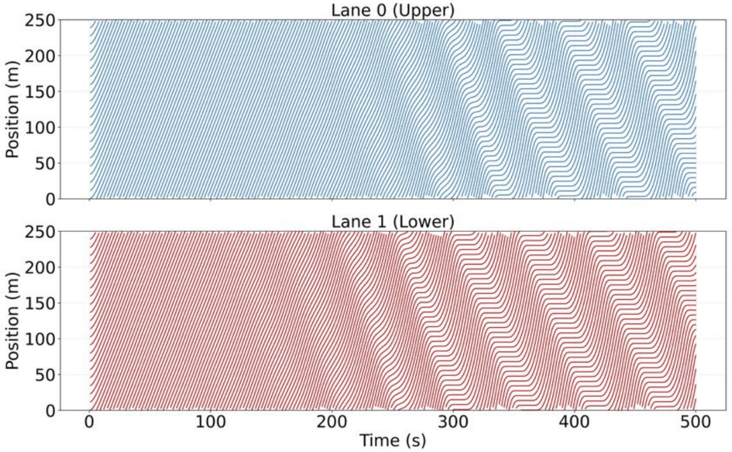
Small braking disturbances travel upstream, creating stop-and-go waves.

Impacts

- Longer travel times and unnecessary acceleration and braking
- Higher collision risk, energy use, and emissions

Stop-and-go wave





Stop-and-go waves are usually treated as a longitudinal-control problem

Existing Methods Stabilize Traffic Longitudinally only

Rule-based Control

Predefined laws regulate acceleration using headway and relative speed.

ACC · CACC · FollowerStopper

Simple, interpretable, and deployable

Limitations

Usually designed for longitudinal one lane

Vahidi & Eskandarian, 2004; van Arem et al., 2006; Wu et al., 2018

Optimization-based Control

Traffic models optimize vehicle inputs under explicit objectives and constraints.

Robust control · H∞ control · MPC

Handles constraints and multiple objectives

Limitations

Computation grow under unpredictable lane-changing dynamics.

Zheng et al., 2020; Feng et al., 2021

Learning-based Control

Policies are learned directly from interactions with mixed traffic.

Deep RL · Multi-agent RL

Complex interactions and adaptive behaviors

Limitations

Difficult to interpret, verify, and deploy.

Wu et al., 2018; Shi et al., 2023; Suriyarachchi et al., 2022

Shared Gap

Longitudinal control **under simplified car-following dynamics without lane changing behaviors**

Human Lane Change → New Cross-lane Disturbance

Motivation: Stabilizing Stop-and-go under Mixed Traffic with Human Lane Changes

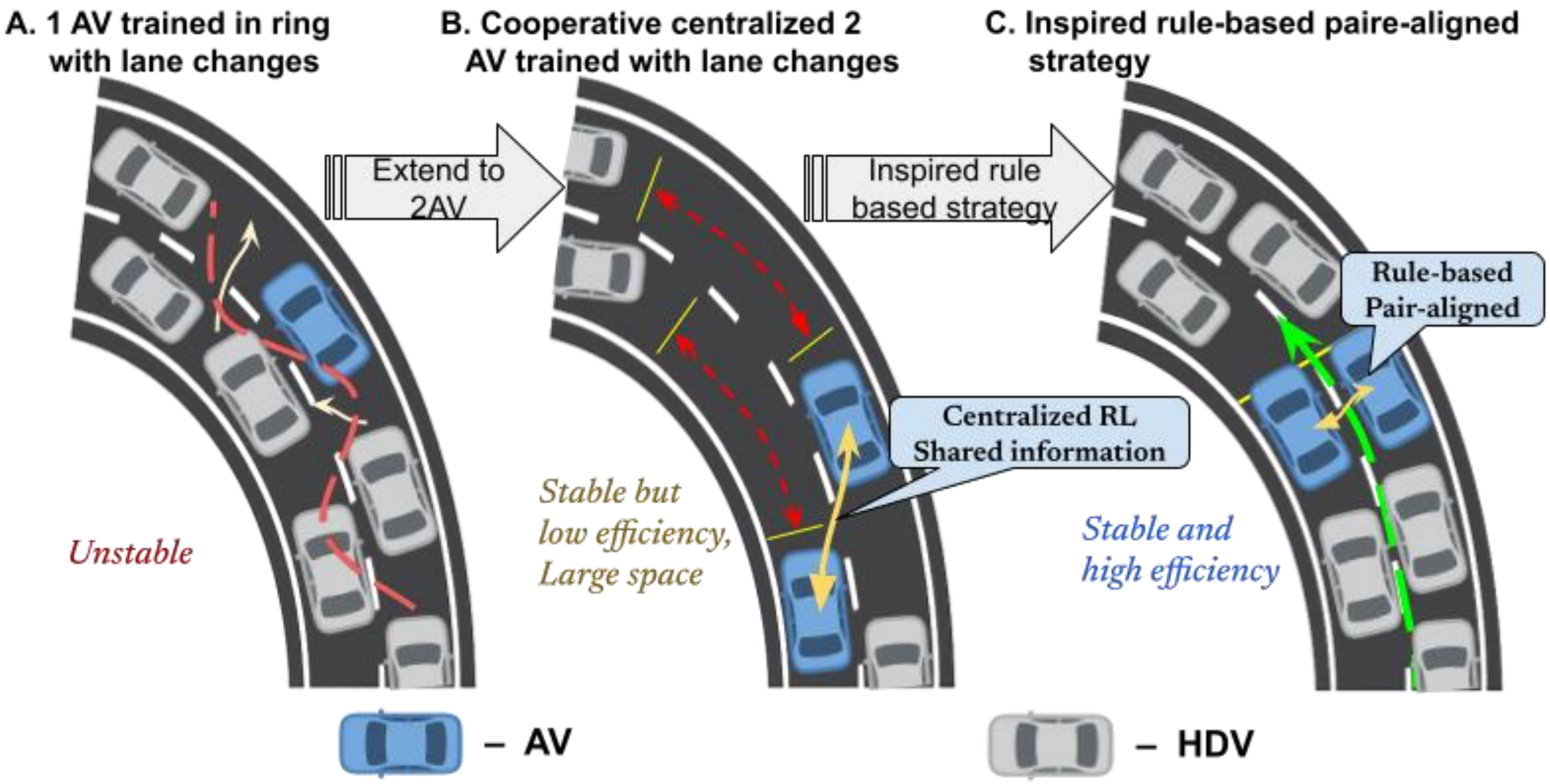
Sources: cited ACC/CACC, MPC, RL literature; Yan et al. (2023)

From Single Lane to Mixed Traffic with Human Lane Changes

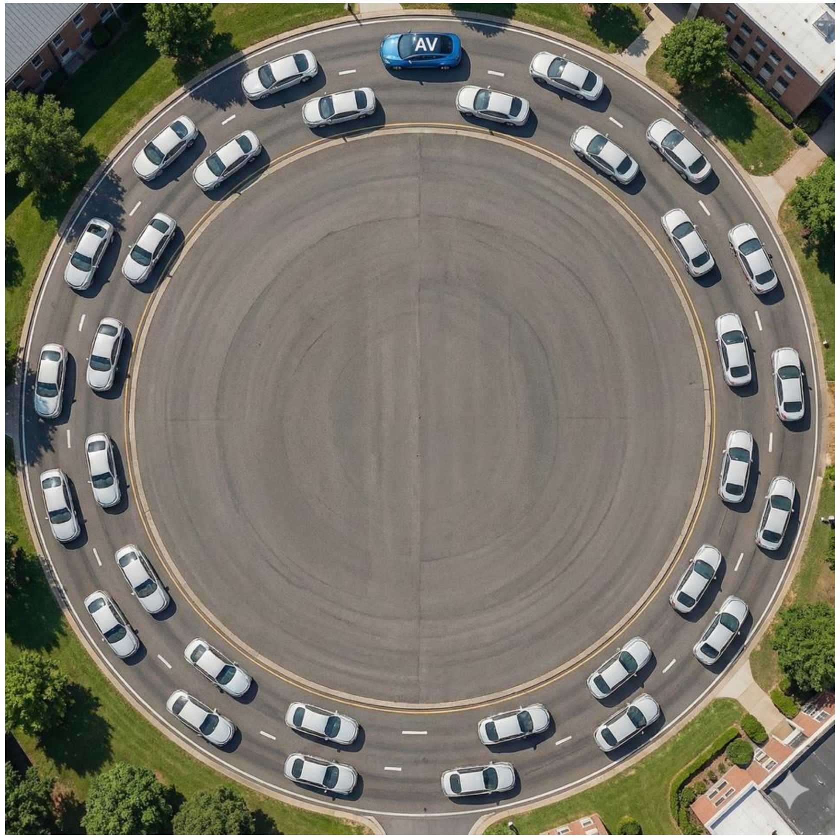
Q1 · Generalization
Can one AV stabilize traffic with human lane changes?

Q2 · Behavior discovery
What coordination emerges when two AVs share information?

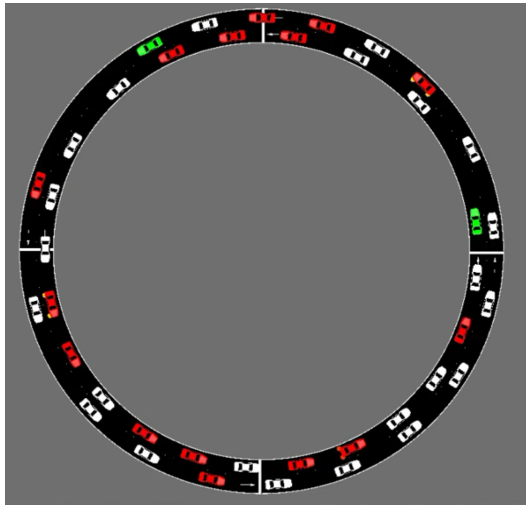
Q3 · Controller design
Can the learned structure become an interpretable controller?



Double-Lane Ring Road with Lane Changing Behavior



- ✓ Two-lane ring road, L = 250 m
- ✓ N_{total} = 44 vehicles
- ✓ SUMO microscopic simulation
- ✓ Flow reinforcement-learning interface
- ✓ IDM longitudinal dynamics model
- ✓ Heterogeneous LC2013 lane-changing behavior
- ✓ Bounded AV acceleration commands $a_t^{\mathrm{acc}} \in [-0.5, 0.5]$



HDVs: IDM + LC2013 with randomized parameter
AVs: learned or rule-based longitudinal control

A warm-up period of 500 seconds is introduced to generate traffic oscillations.

Formulation as a finite-horizon MDP

Formulation

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r, \rho_0, \gamma, H)$$

$$\max_{\pi} J(\pi) = \mathbb{E}_{\pi,\mathcal{T}}\left[\sum_{t=0}^{H-1} \gamma^t\, r(s_t, a_t, s_{t+1})\right]$$

$$s_0 \sim \rho_0, \quad a_t \sim \pi(\cdot|\, s_t), \quad s_{t+1} \sim \mathcal{T}(\cdot|\, s_t, a_t)$$

Traffic Interpretation

$\mathcal{S}$	**Traffic observations** Local traffic and AV coordination states
$\mathcal{A}$	**Control decisions** Acceleration and lane-change decisions
$\mathcal{T}$	**Traffic evolution** Stochastic SUMO traffic evolution
r	**Control objective** Efficiency, synchronization, and smooth
ρ_0, γ, H	**Episode settings** Initial state, discounting, and horizon

The vehicular control problems considered are naturally viewed as infinite-horizon MDPs ($H \rightarrow \infty$).
In practice, we approximate this by **choosing a large but finite horizon H.**

The single AV observes both lanes.

➢ **Observation $\mathcal{S}$**

$$o_t = \left[\frac{v_t^{\mathrm{AV}}}{v_{\max}}, \mathbf{z}_t^{(0)}, \mathbf{z}_t^{(1)}\right] \in \mathbb{R}^{11}$$

$$\mathbf{z}_t^{(\ell)} = \left[\mathbf{1}_{\{\ell=\ell_t\}}, \frac{d_{f,t}^{\ell}}{L}, \frac{v_{f,t}^{\ell}}{v_{\max}}, \frac{d_{b,t}^{\ell}}{L}, \frac{v_{b,t}^{\ell}}{v_{\max}}\right]$$

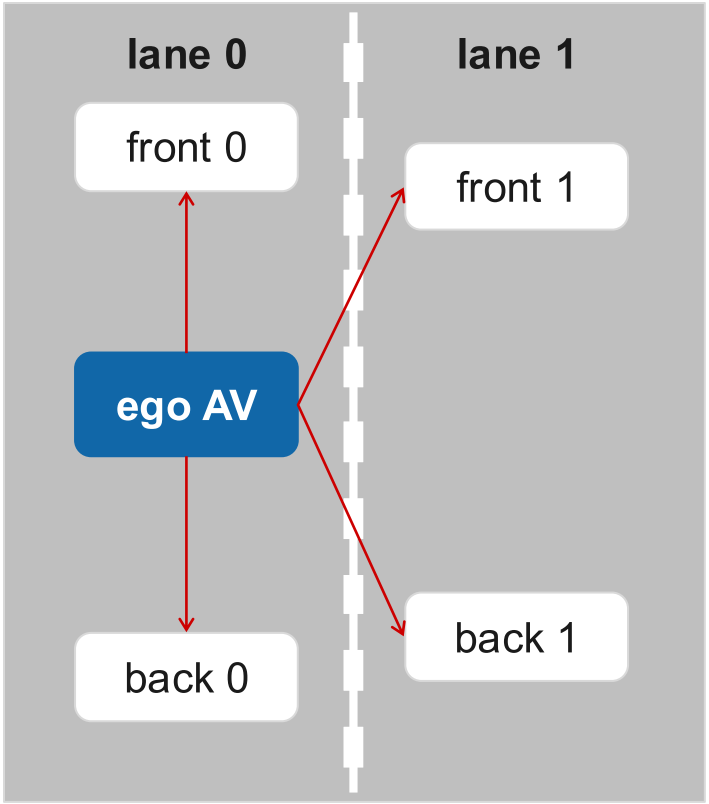
➢ **Action $\mathcal{A}$**

$$a_t = \left[a_t^{\mathrm{acc}}, a_t^{\mathrm{lc,raw}}\right]$$

$$a_t^{\mathrm{acc}} \in [-0.5,\, 0.5] \quad a_t^{\mathrm{lc}} \in \{-1,\, 0,\, +1\}$$

➢ **Reward r**

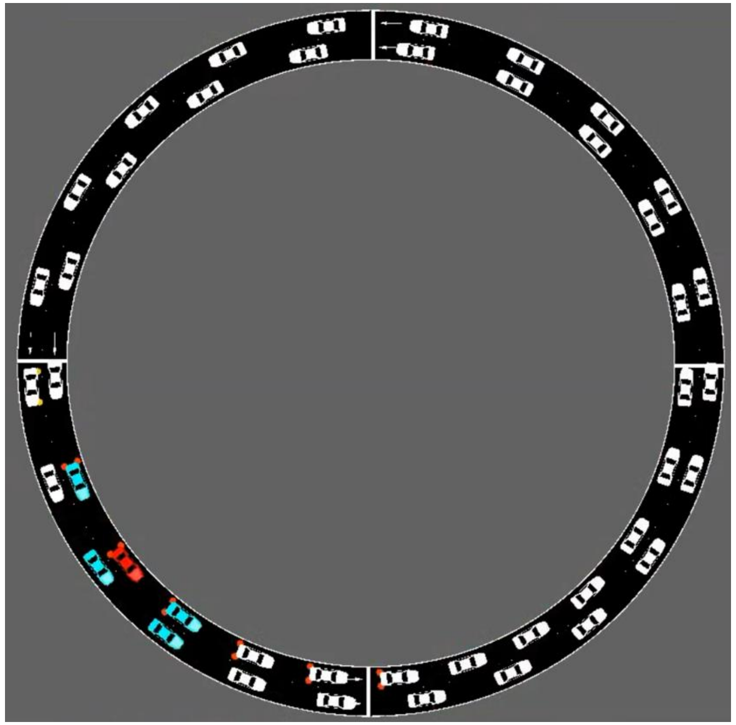
$$r_t = \frac{1}{N}\sum_{j=1}^{N} v_{j,t}$$



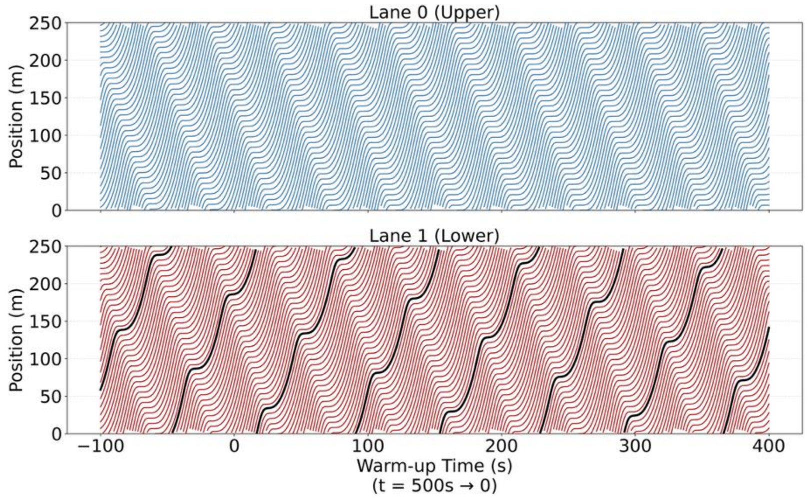
Results and Analysis

Single AV can not stabilize stop wave under lane changes.

- The AV learns to close the gap to its leader.
- Large gap invites lane-changing HDVs, causing oscillations.
- Short gap suppressed lane-changing attempts.
- It reduces nearby merging but cannot stabilize the lane.



✓ A locally optimal but globally insufficient strategy.
✓ Suppressing unnecessary lane-changing is key to stability.



Cooperative AVs to suppress **lane-changing** and enhance stability.

Two Avs are jointly governed by a shared RL policy

➢ Centralized Observation $\mathcal{S}$

$$o_t = \left[o_t^1, o_t^2, d_{\mathrm{AV},t}, \Delta v_{\mathrm{AV},t}\right] \in \mathbb{R}^{12}$$

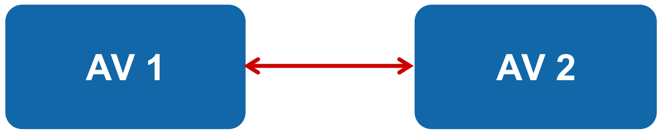
$$o_t^i = \left[\frac{v_t^i}{v_{\max}}, \frac{v_{i,t}^{\mathrm{lead}}}{v_{\max}}, \frac{v_{i,t}^{\mathrm{foll}}}{v_{\max}}, \frac{d_{i,t}^{\mathrm{lead}}}{L}, \frac{d_{i,t}^{\mathrm{foll}}}{L}\right], \quad i \in \{1,2\}$$

$$\left[d_{\mathrm{AV},t}, \Delta v_{\mathrm{AV},t}\right] = \left[\frac{x_t^1 - x_t^2}{L}, \frac{v_t^1 - v_t^2}{v_{\max}}\right] \in \mathbb{R}^2$$

➢ Longitude Action $\mathcal{A}$

$$a_t = \begin{bmatrix} a_{1,t} \\ a_{2,t} \end{bmatrix} \in [-0.5, 0.5]^2$$

- ✓ Centralized Observation
- ✓ Two longitudinal control inputs
- ✓ AVs remain in assigned lanes
- ✓ Relative position and velocity



Partner State

Cooperative Reward Design: Reduce gap to suppress Lane Change

$$r_t = r_{\text{pos},t} + r_{\text{sync},t} + r_{\text{speed},t} + r_{\text{smooth},t}$$

$$r_{\text{pos},t} = -w_{\text{pos}} \left| \frac{x_t^1 - x_t^2}{L} \right|$$

Position → alignment

$$r_{\text{sync},t} = -w_{\text{sync}} \left| \frac{v_t^1 - v_t^2}{v_{\max}} \right|$$

Synchronization → coherent motion

$$r_{\text{speed},t} = w_{\text{speed}} \left(\frac{v_t^1 + v_t^2}{2v_{\max}} \right)^2$$

Speed → traffic efficiency

$$r_{\text{smooth},t} = -w_{\text{smooth}} \left((a_t^1)^2 + (a_t^2)^2 \right)$$

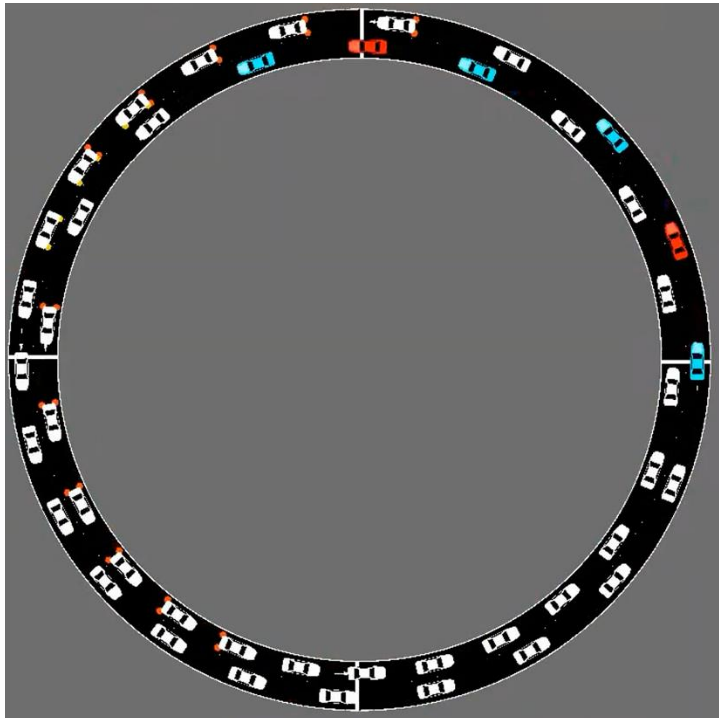
Smoothness → feasible acceleration

Results and Analysis

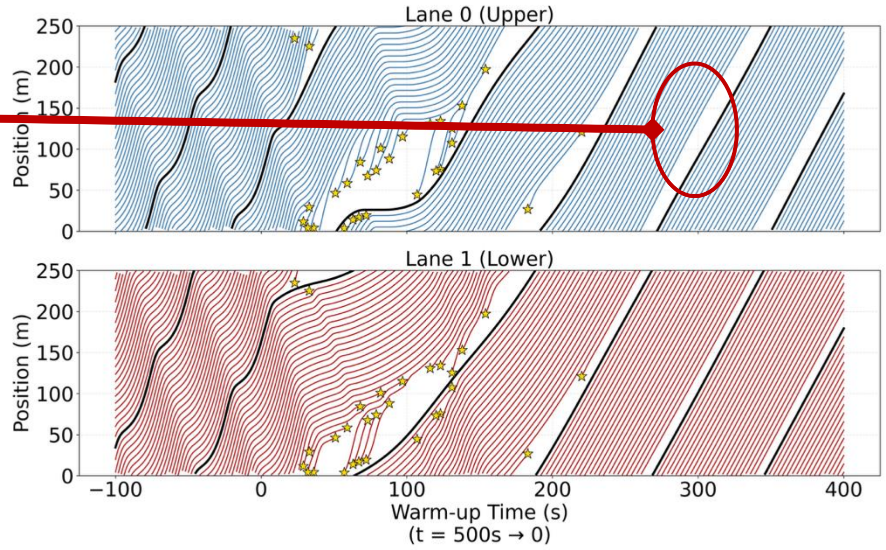
Suppressed Lane-changes and stabilized the wave.

Behaved as one virtual longitudinal lane.

- The leading AV slows down. The trailing AV catch up.
- Close the gap to prevent human lane changes.
- Two Avs align longitudinally and paired together.

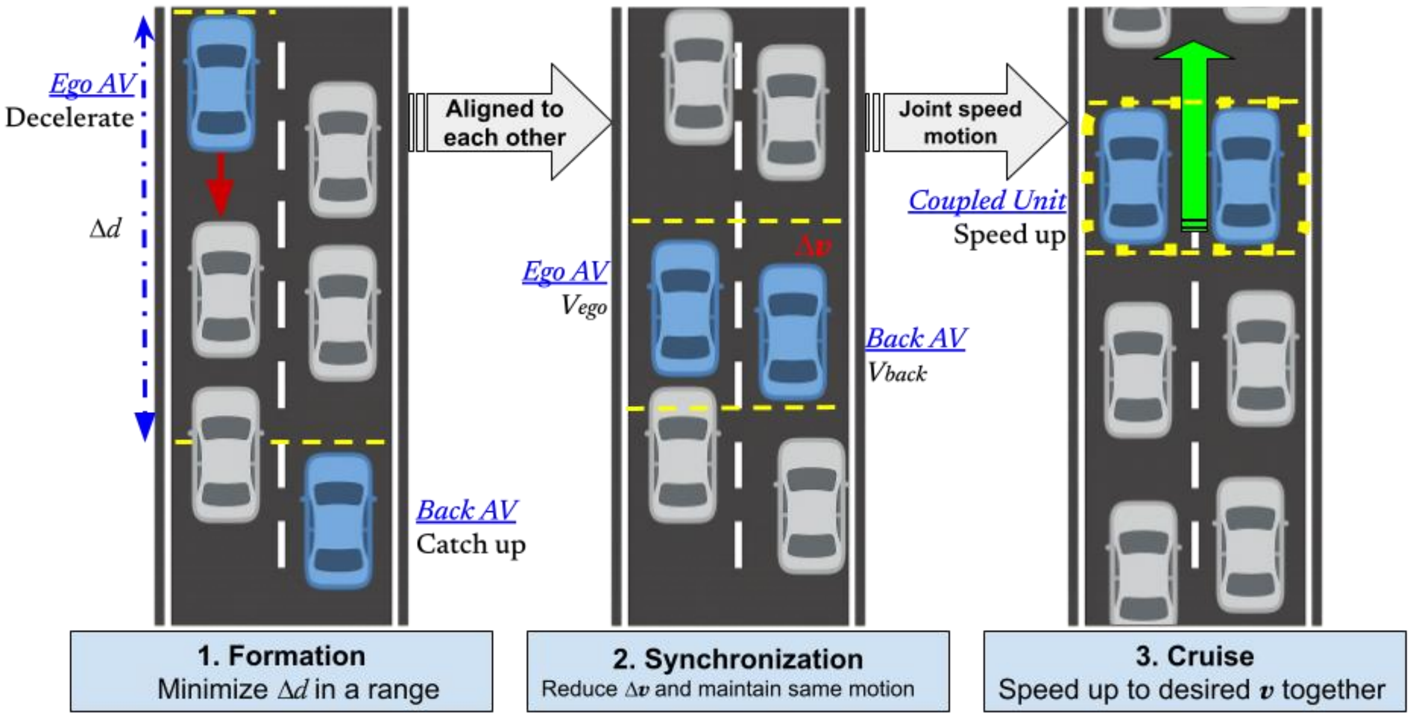


x Considerable space ahead of them.
x Indicating potential for improving average speed and overall traffic efficiency.



Suggests a **pairing control mechanism**, lower **computational cost and greater interpretability**

PARC Encodes the Learned Mechanism.



Algorithm 1 Pair-Aligned Control Strategy

Require: Ego state (x, v); partner state (x_p, v_p); control gains $(k_{\text{pair}}, k_{\text{sync}}, k_v)$;
Ensure: Acceleration command a

1: $d_p \leftarrow (x_p - x)$ ▷ Signed periodic displacement
2: **if** $|d_p| > d_{th}$ **then**
3: $\quad a \leftarrow k_{\text{pair}}\, d_p$ ▷ Formation: spatial alignment
4: **else if** $|v_p - v| > \delta_v$ **then**
5: $\quad a \leftarrow k_{\text{sync}}(v_p - v)$ ▷ Synchronization
6: **else**
7: $\quad a \leftarrow k_v(v^\star - v)$ ▷ Cruise: joint steady-state
8: **end if**
9: $a \leftarrow \text{clip}(a, -a_{\max}, a_{\max})$
10: **return** a

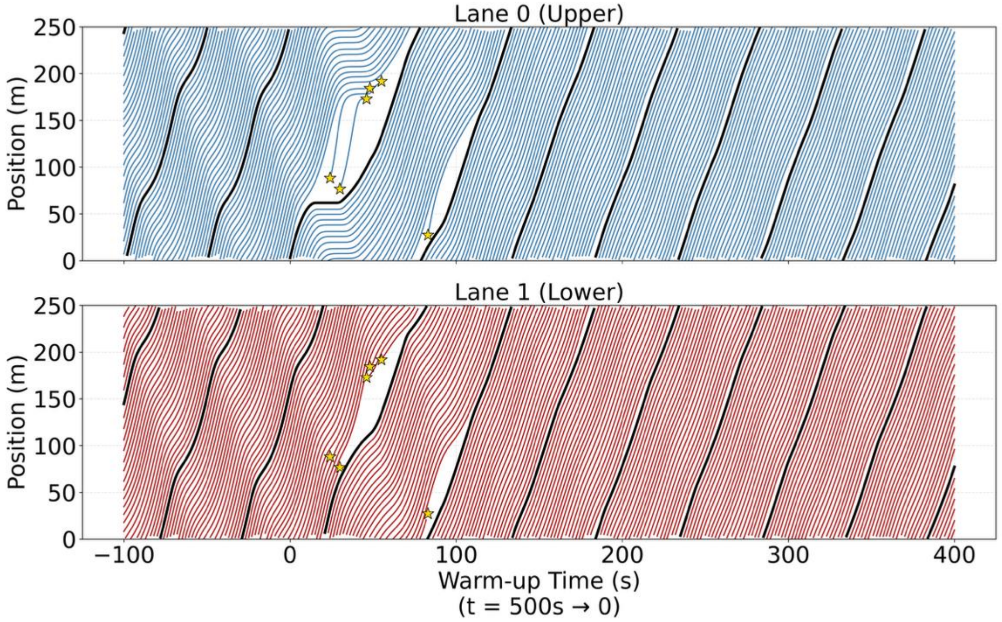
$$s_{\text{eq}}^{\max} = (s_0 + v^\star \tau)\left[1 - \left(\frac{v^\star}{v_0}\right)^{\gamma}\right]^{-\frac{1}{2}} \qquad s_{\text{eq}}^{\max} = \frac{L - N s_{\min}}{N - 1}$$

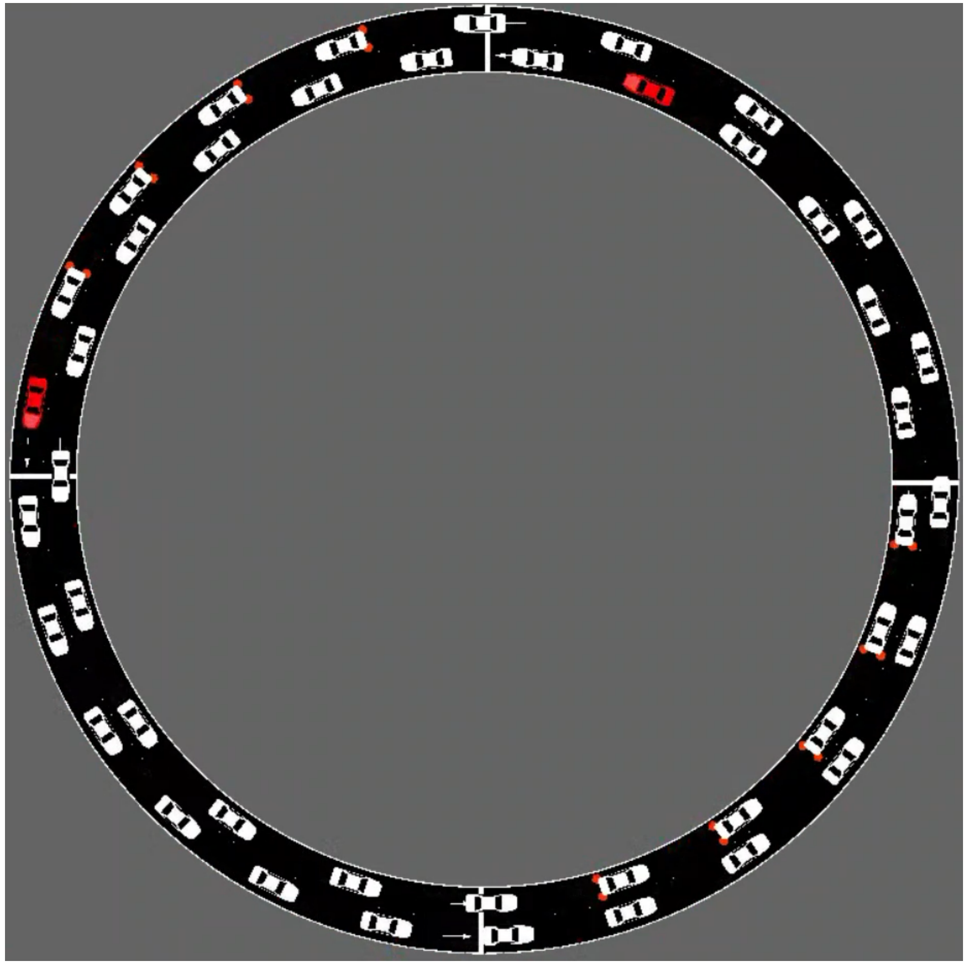
The cruise speed is related to lane density, safety spacing and car following model.
—not only the road speed limit.

Results and Analysis

After alignment: nearly parallel trajectories.
Paired structure improve stability and average speed.

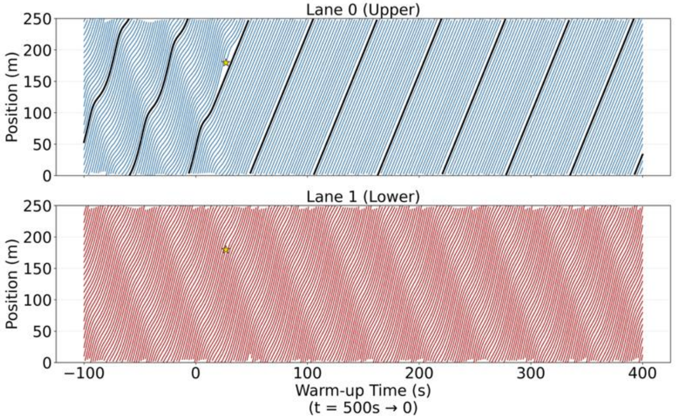
- Complete the gap quickly, fewer merging chances.
- Couples the two lanes into a single virtual lane.





Time–space Trajectories under Five Control Strategies

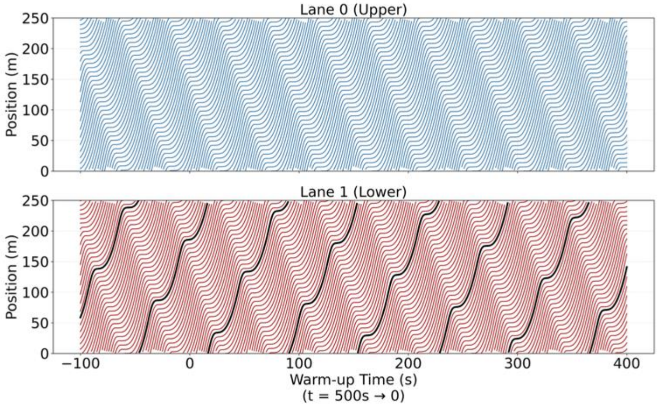
➢ Compare with rule-based controller from *Wu et al.(2018)* in double-lane environment.



1AV-SLC

Partial stabilization

Single-AV rule-based single lane controller *(Wu et al., 2018)*, extracted from RL.



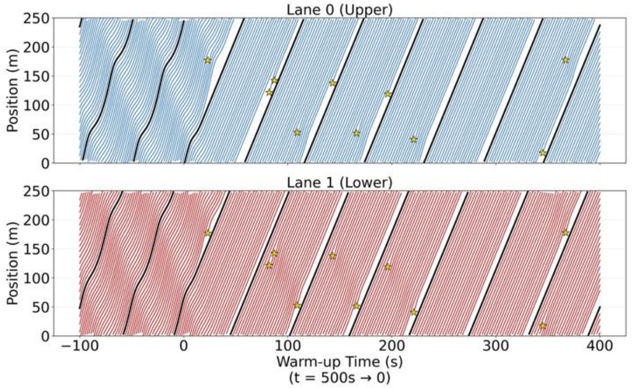
RL-1AV

Unstable

Single-AV RL controller

2AV-SLC

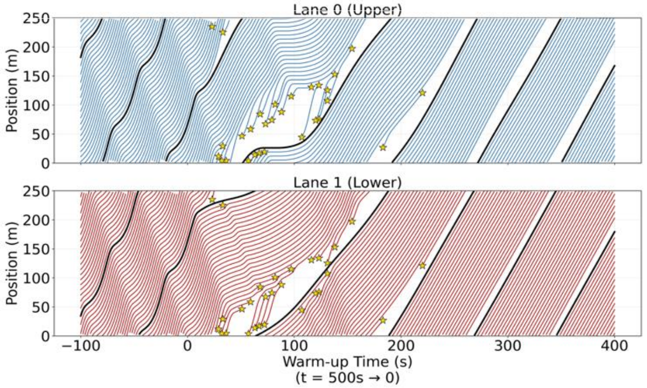
Two independent SLC controllers, applied separately to both lanes. *(Wu et al., 2018)*



Stable, few disturbances

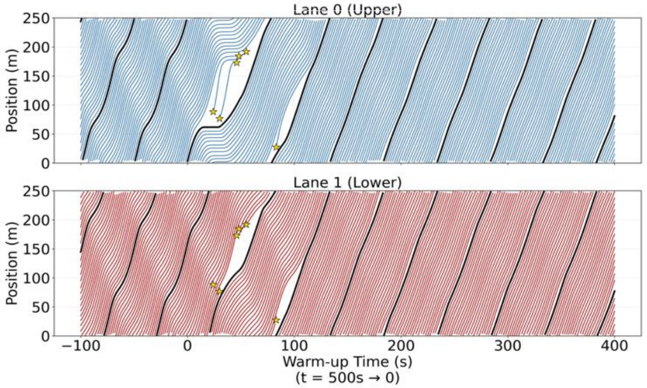
RL-Coop-2AV

Cooperative two-AV RL controller



Stable but slow

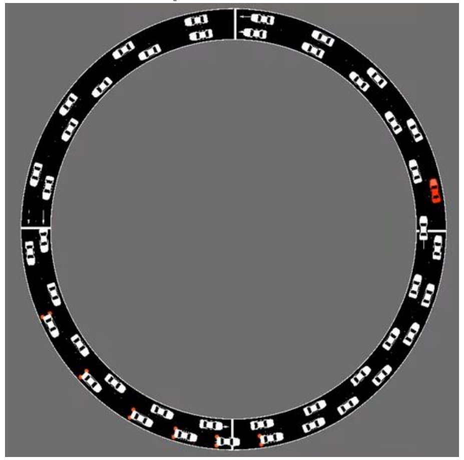
PARC Proposed



Stable and efficient

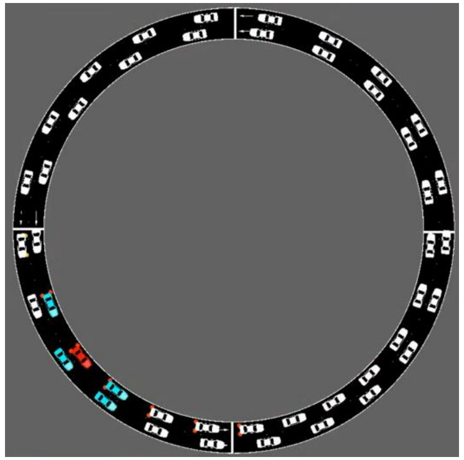
Dynamic Velocity Comparison

1AV-SLC *(Wu et al., 2018)*



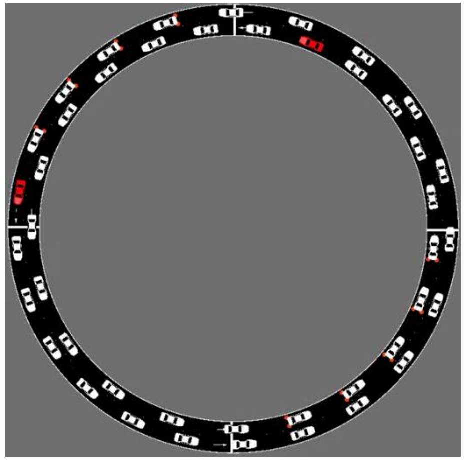
Local stabilization only

RL-1AV



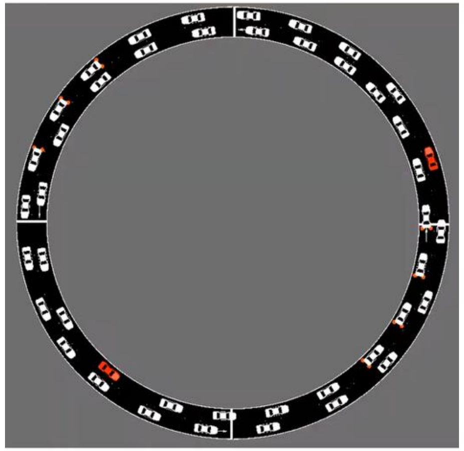
Unstable

PARC — Proposed



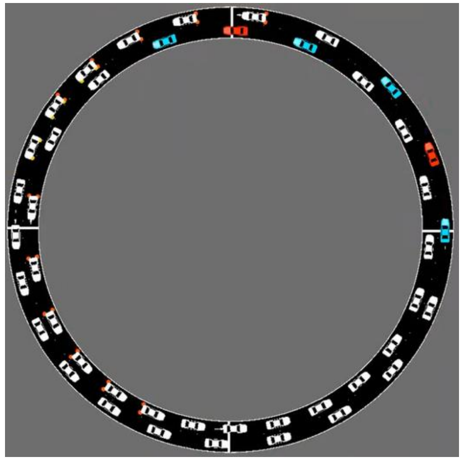
Stable · higher speed

2AV-SLC *(Wu et al., 2018)*



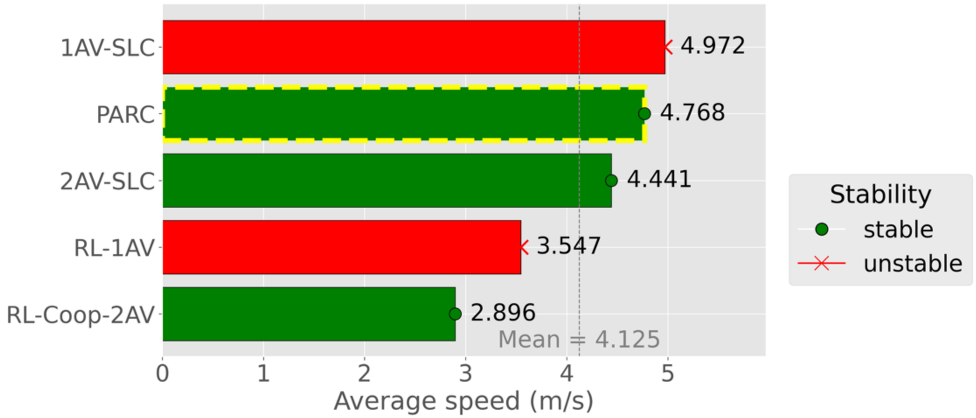
Stable, few disturbances

RL-Coop-2AV



Stable, low speed

Average performance comparison

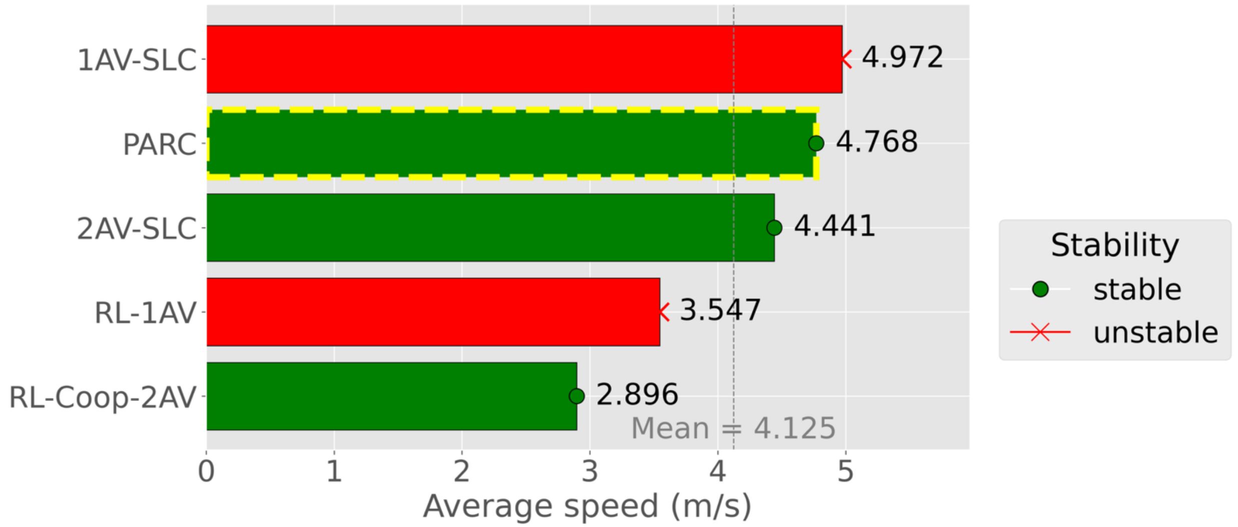


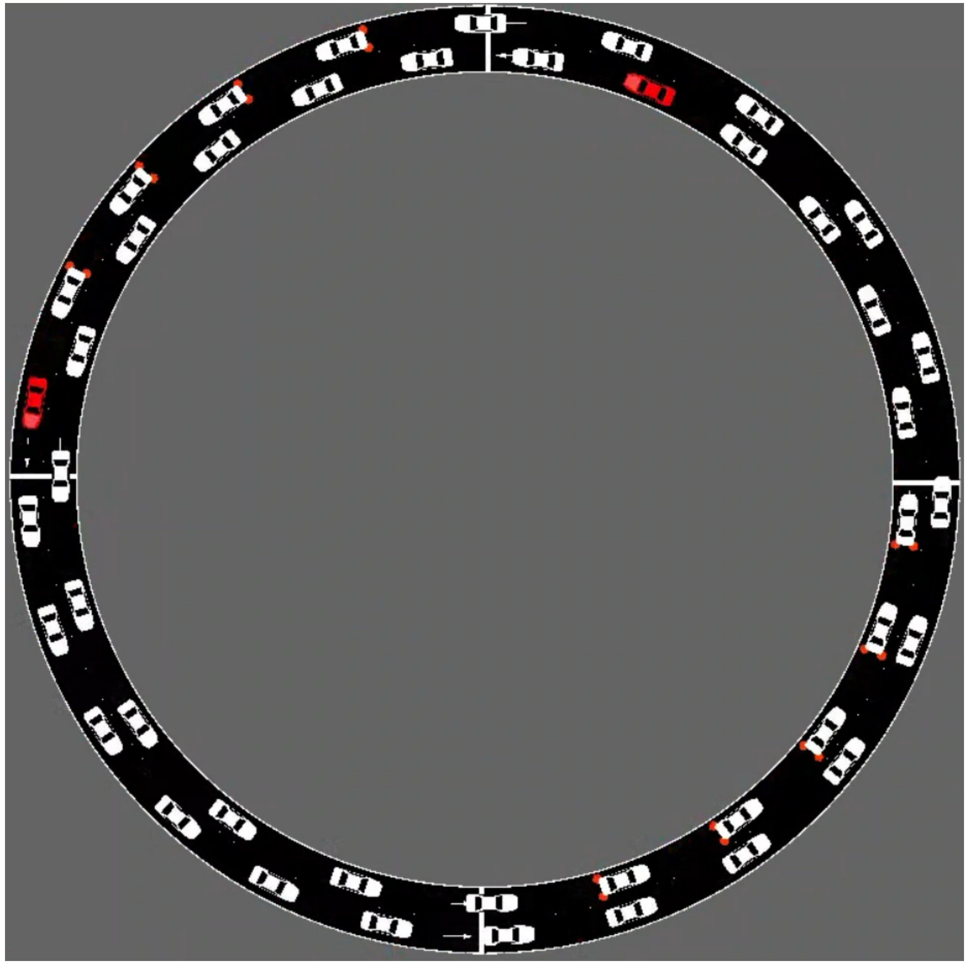
7. Conclusion

- One AV is **insufficient** in lane-changing environment.
- Cooperative design identifies **cross-lane pairing** methods.
- PARC improves **stability and average speed.**
- It is easy to **implement** and **interpretable**.

$$4.441\ \mathrm{m/s} \rightarrow 4.768\ \mathrm{m/s}$$

(4.768 − 4.441) / 4.441 × 100% = 7.4%





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